



Cairo University
 Faculty of Engineering
 Credit Hours System
Communications and Computer Engineering Department

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Course Code EECS 304
 Course Title Control I
 Course Instructors Dr. Ragia Badr - Dr. Nasr Antar
 Course TA Eng. Hassan El-Menier

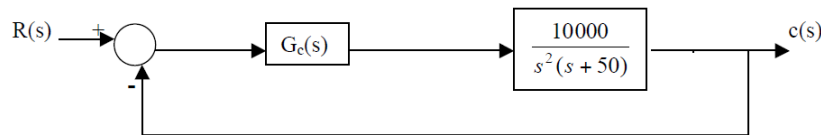
Sheet 7: Control Design and Compensation

Question (1):

The block diagram of a type 2 system with a cascade controller $G_c(s)$ is shown in figure. The objective is to design a controller $G_c(s)$ so that:

- Maximum overshoot $< 10\%$
- Settling time < 1

Design the controller that satisfies the performance specifications given above.



Question (2):

Consider a unity feedback controller system whose feed-forward transfer function is:

$$G(s) = \frac{2}{s(s+2)(s+8)}$$

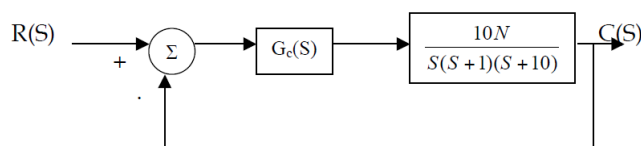
- Find the static error coefficients and the error as a function of time for unit ramp input.
- Design a compensator so that the closed loop dominant poles are located at $s = -1 \pm j2\sqrt{3}$

Question (3):

The following figure shows the block diagram of the liquid level control system. The liquid level is represented by $c(t)$, and N denotes the number of inlets. It is desired that $N = 20$. Design a controller $G_c(s)$ such that:

- The overshoot is zero.
- The tank is filled to the reference level within 2.5 sec.

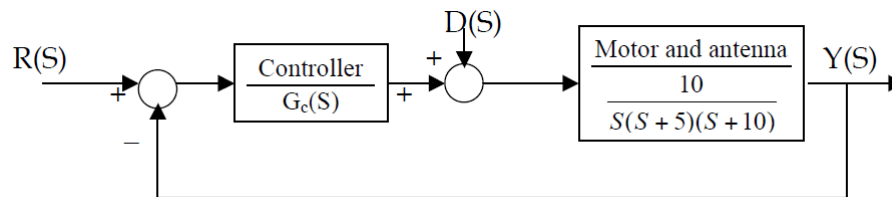
What is the effect of the controller on the steady state response? And why?



Question (4):

A large antenna is used to receive satellite signals and must accurately track the satellite as it moves across the sky. The control system uses an armature controlled motor and a controller to be selected as shown in figure. The system specifications require an overshoot to a step input less than or equal 10% with a settling time less than or equal 2 seconds.

- Design a controller $G_c(s)$ to achieve the required specifications and plot the resulting response.
- Determine the effect of the disturbance $[D(S) = Q/S]$ on the steady state output.

**Question (5):**

A control system with a type 0 process and PID controller is shown in figure. Design the parameters of the PID controller so that the following specifications are satisfied:

- Settling time ≤ 0.2
- Max. Overshoot ≤ 5 percent.
- Steady state error due to unit step disturbance = zero.

